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# Extraction, characterization and biological activity of citrus flavonoids

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**Abstract:** Citrus is one of the largest and most popular fruit crops commercially grown across the globe. It is not only important in terms of economy but is also popular for its nutritional benefits to human and farm animals. Citrus is available in several varieties, all with attractive colors. It is consumed either fresh or in processed form. After processing, approximately 50% of the fruit remains unconsumed and discarded as waste. The latter includes fruit pith residue, peels and seeds. Direct disposal of these wastes to the environment causes serious problems as these contain bioactive compounds. Release of these bioactive compounds to the open landfills cause bad odor and spread of diseases, and disposal to water bodies or seepage to the underground water table deteriorates water quality and harms aquatic life. In this regard, a number of research are being focused on the development of better reuse methods to obtain value-added phytochemicals as well as for safe disposal. The important phytochemicals obtained from citrus include essential oils, flavonoids, citric acid, pectin, etc., which have now become popular topics in industrial research, food and synthetic chemistry. The present article reviews recent advances in exploring the effects of flavonoids obtained from citrus wastes, the extraction procedure and their usage in view of various health benefits.

**Keywords:** citrus; flavonoids; hesperidin; naringenin; polyphenols.

## 1 Introduction

Citrus fruits have been known to humankind since ancient times for health benefits due to their nutrient contents and secondary metabolites, such as vitamin C and B complex, phenolics, flavonoids, pectin, etc. Vitamin C is important for fighting infections and building immune system. It helps in the healing of the mucosal lining by stimulating procollagen formation and subsequent synthesis of the same (Sood et al. 2009). Various parts of citrus fruits, such as peels, pulp, seeds, juice, rind, etc., are popular in traditional Indian Ayurveda and Siddha medicines as well as Oriental medicines, in fresh or dried form. Citrus fruits have curative effects on sore throat, cough, asthma, thirst, hiccup, earache, nausea, indigestion and vomiting. These are also potential anti-scorbutic, stomachic tonic, stimulant expellant of poison, analgesic and help in removing fetid breath. Distilled water extract of the fruit is used as a sedative, and the extract of the fruit and seeds together is useful in palpitation and is used in cardiac tonics (Peter et al. 2008, Nagaraju et al. 2012).

Citrus belongs to the Rutaceae family. It comprises 140 genera and 1300 species. It is believed to have originated in the Himalayan foothills of Northern India, Burma, Southern China and Southeast Asia. The important citrus species for commercial cultivation are *Citrus maxima* (pomelo), *Citrus medica* (citron), *Citrus micrantha* (papeda), *Citrus reticulata* (mandarin orange), etc. The main native citrus species selected by the humankind for cultivation during early civilization were *C. medica* (citron) from India; *C. maxima* (pomelo or shaddock) from the Malay archipelago; *Citrus japonica* (kumquats), *Citrus junos* (yuzu) and *Poncirus trifoliata* (Japanese/Korean bitter orange) from East Asia, namely Japan, Korea and China; *Citrus crenatifolia* (heennaran), which can be found only in Sri Lanka; *Citrus mangshanensis* from the Hunan Province in China; *Citrus platymamma* (byeonggyul); *C. reticulata* (mandarin orange) from China; *Citrus trifoliata* (trifoliate bitter orange) from Korea, China and Japan; *Citrus sudachi* (Sudachipapeda); *Citrus australasica* (Australian finger lime, which is native to Australia); etc. With improving human lifestyle and increasing demands for citrus fruits, different hybrid varieties were developed from time to time by researchers, farmers and horticulturists. The main hybrid varieties are sweet orange (*Citrus sinensis*), tangerine or mandarin (*C. reticulata*), grapefruit (*Citrus vitis*), lime (*Citrus aurentifolia*), lemon (*Citrus limonum*), etc. The commercially important citrus varieties are shown in Figure 1. Currently, the availability of citrus is not only limited to Southeast Asia, but they

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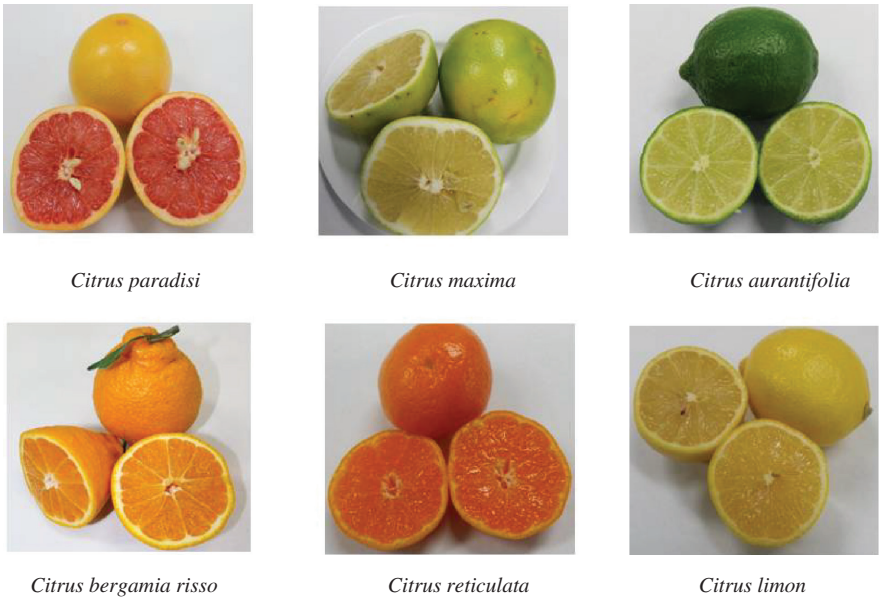



**Figure 1:** Important and most popular citrus varieties.

are also widely grown in many tropical and sub-tropical countries worldwide. There are currently 140 citrus-producing countries, and the annual citrus production worldwide amounts over 70 million tons (Sharma et al. 2017). Of the total citrus production, ~40–50% is utilized in the processing and manufacturing of commercial products, which include juice, jams, marmalades, jellies, flavoring agents, beverages and health drinks, etc.

## 2 Citrus wastes and its effects on the environment

Citrus processing industries produce huge amounts of waste annually, which mounts to over 40 million tons worldwide. The waste contains almost 50% of the original fruit mass, of which ~40–55% are peels, ~30–35% internal tissues and ~10% seeds. The ratio of the waste composition depends upon species, variety and climatic conditions and cultivation. The citrus waste mainly contains flavedo or exocarp, albedo and pith residue. It contains high amounts of soluble sugars, fibers, endocarp residual membranes containing some amounts of juice and moisture (>85%). The semi-solid waste therefore is prone to bacterial growth and fermentation. The peels are rich in essential oils, especially limonene (30–80%), possessing antibacterial properties. This renders citrus wastes inappropriate for direct disposals to landfills or dumping grounds. Direct disposal of untreated citrus wastes adversely affects the natural and beneficial flora present in the soil and aquatic bodies. Particularly, oils present in citrus peels ooze from the oil glands, float on the water surface and block the passage of oxygen from the atmosphere into the water, harming aquatic life. In addition, citrus wastes are very difficult to dry using common conventional methods. It is therefore very important to develop better disposal methods to save the environment.

A number of commercially important value-added compounds can be extracted from citrus wastes and utilized in food and processing industries, pharmaceutical companies, domestic usage, etc. The major value-added compounds are essential oils (mainly D-limonene), flavonoids (hesperidin, naringenin, neo-hesperidin), carotenoids (lutein, β-carotene, lycopene, zeaxanthin), limonoids (limonin, normilin, limonoic acid), phenolics (coumarin, phenolic acid, phloroglucinol), organic acids (citric acid, maleic acid, succinic acid), vitamins (ascorbic acid, niacin, riboflavin), carbohydrates, pectins and enzymes (pectinesterase, phosphatase, peroxidase) (Sharma et al. 2017).

## 3 Citrus flavonoids

Flavonoids are secondary metabolites naturally occurring in plants and possessing significant biological properties. These play an important role in protecting plants against ultraviolet (UV) radiation. These are bitter, which prevents herbivores from grazing the crop. In addition, these also

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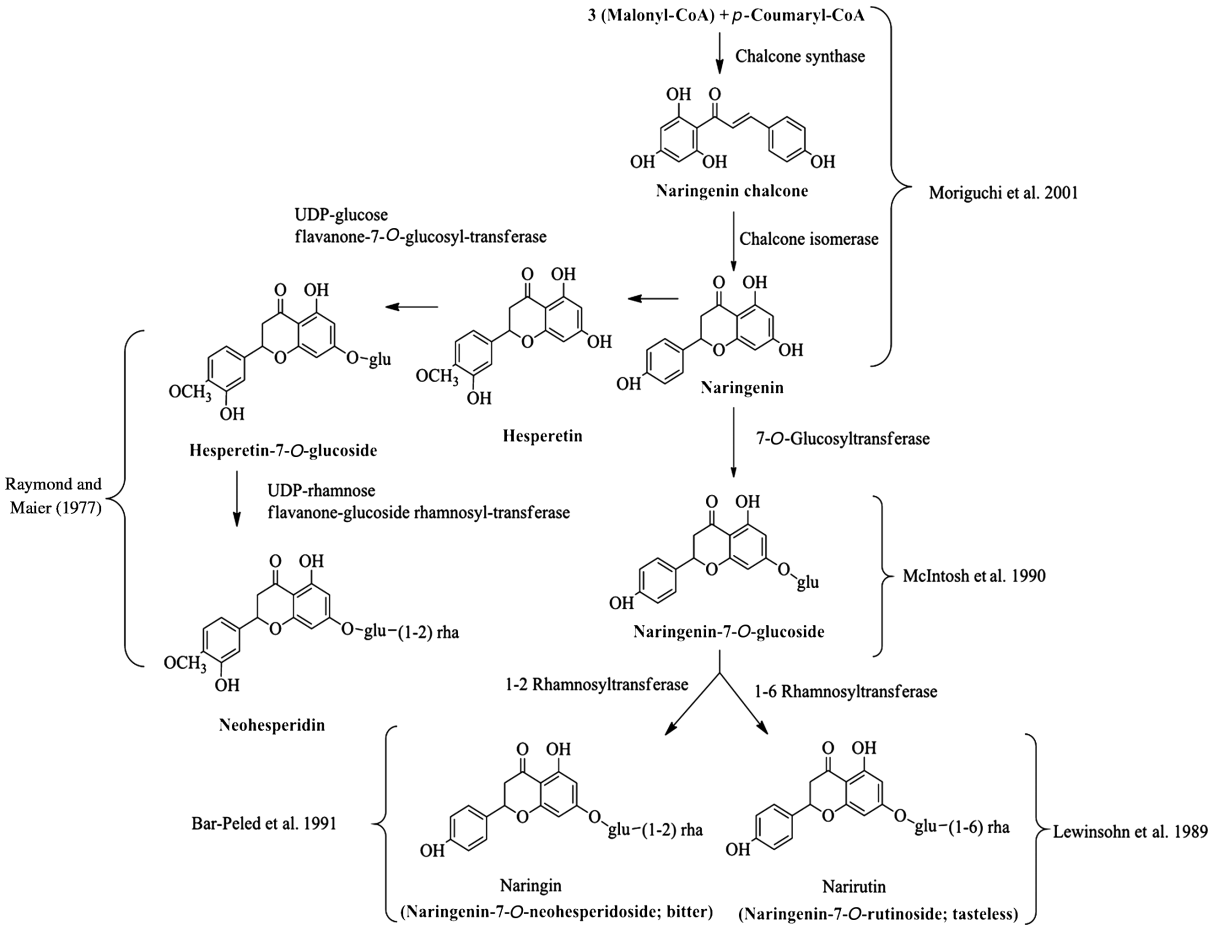



**Figure 2:** Biosynthesis pathway of citrus flavanone.

possess antibacterial properties and protect the plant from pathogens and infections (Heim et al. 2002). Flavonoids are responsible for coloration of flowers and fruits. These are found in many varieties of foods, *viz*. fruits, vegetables, nuts, cocoa, seeds, soy, beverages, coffee, tea, wine etc. and thus constitute a significant portion of human diet. Thus far, hundreds of flavonoids have been identified and reported. The basic structure comprises 15 carbon atoms. Flavonoids basically belong to the polyphenolic group, which is characterized by two phenolic rings bearing one or more hydroxyl group(s) connected by a 3-carbon chain. Therefore, the basic structure of flavonoids is referred as C6-C3-C6. All flavonoids arise from a common initial reaction catalyzed by chalcone synthase enzyme. The chalcone is rapidly converted into phenylbenzopyran followed by further modification, leading to the formation of flavones, isoflavones, flavonols or anthocyanins. Flavonoids can exist in free aglycone form but have been observed to often bind with glycosides (most commonly glucose). The later are water soluble (Croft 1998). The gut microflora present in the animal intestines breaks down flavonoid glycosides into aglycone, which is easily absorbed. Upon absorption, flavonoids may enter into various metabolic processes, where hydroxyl groups are added to their basic structure, or they can be methylated, glucuronidated or conjugated to sulfates, mainly regulated by the liver (Croft 1998). Flavonoids have been divided into many classes on the basis of their structural variation and degree of oxidation. Theseare flavones, isoflavones, flavanones, flavonols, flavanols and anthocyanidins (Mulvihill and Huff 2010).

Citruses are of great interest due to large amounts of flavonoid glycosides which are accumulated in the fruits and leaves during flavonoid synthesis. In the flavonoid biosynthetic pathway (Figure 2), aglycones are the intermediates for flavonoid glycosides. The flavonoid pathway in the citrus has been found similar to that studied in other species. Chalcone synthase is a key enzyme in the flavonoid pathway. Chalcone synthase catalyzes the condensation of *p*-coumaryl-CoA with three molecules of malonyl-CoA yielding naringenin-chalcone (Moriguchi et al. 2001). Flavanone aglycones are converted into their

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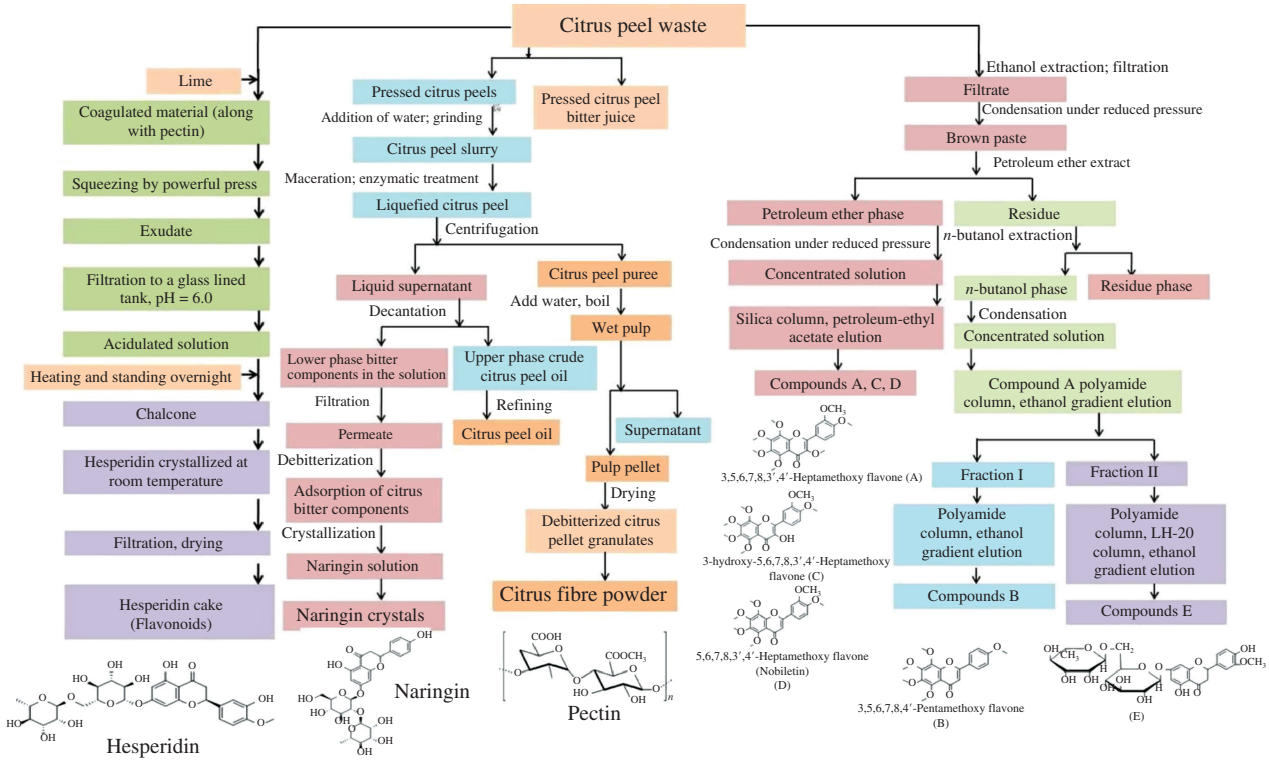



**Figure 3:** Schematic representation of the recovery of bioactive compounds from citrus waste (Agricultural Research Service 1956, Nafisi-Movaghar et al. 2013, Shan 2016).

corresponding glucosides and rhamnoglucosides by UDP glucose flavanone-7-*O*-glucosyltransferase and UDP-rhamnose flavanone glucoside rhamnosyltransferase (McIntosh et al. 1990). The formation of neohesperetin was reported by Raymond and Maier (1977).

Flavonoid concentration in the citrus depends on the age of the plant. Plant tissues undergoing prominent cell divisions possess the highest levels of flavonoids (Castillo et al. 1993). Citrus fruits (e.g. oranges, lemons, grapefruit, mandarins, limes, pomelos, bergamots) represent a significant source of many types of flavonoids. However, citrus peels are also a primary source of molasses, cold-pressed oils, pectin and limonoids. Figure 3 represents the recovery of bioactive compounds from citrus waste. The flowchart displays the important steps involved in the extraction, purification and isolation of main flavonoids, e.g. hesperidin, naringin, nobiletin and other flavone derivatives. During the extraction of the flavonoids from citrus wastes, pectin is obtained as an important co-product.

In literature, the distribution of citrus flavonoids has been reported differently by different research groups. The classification of the citrus flavonoids and the chemical structures of major flavonoids are shown in Figure 4. According to He et al. (1997), citrus flavonoids are composed of three major subgroups, which include flavanones (mainly di- and tri-*O*-glycosides), flavone glycosides (mainly di- and tri-*O*-glycosides and *C*-glycosides) and polymethoxyflavones. More than 60 types of flavonoids are identified in citrus fruits which are classified into five main classes, *viz*. flavones, flavanones, flavonols, flavans and anthocyanins (anthocyanins are found only in blood oranges) (Horowitz and Gentili 1979). According to the molecular structures, flavonoids are divided into six major classes: flavones, flavanones, flavonols, isoflavones, anthocyanidins and flavanols (or catechins) (Peterson et al. 1998). However, in 2013, Di Donna et al. (2013) reported on the flavonoids classified in to two major classes, namely polymethoxylated flavones and glycosylated flavanones. Flavanones are observed to be accumulated in greater quantities in all the citrus varieties compared to flavones. Citrus flavanones are present in glycoside or aglycone forms, but the diglycoside form of flavanones confers the typical taste of citrus fruits (Macheix et al. 1990). Among the aglycone forms, naringenin and hesperetin are the most important flavanones. The citrus *O*-glycosides are present mostly in the form of rutinosides or neohesperidosides (Gionfriddo et al. 1996). Neohesperidosides, such as naringin, neohesperidin and neoeriocitrin, consist of a basic flavanone structure with

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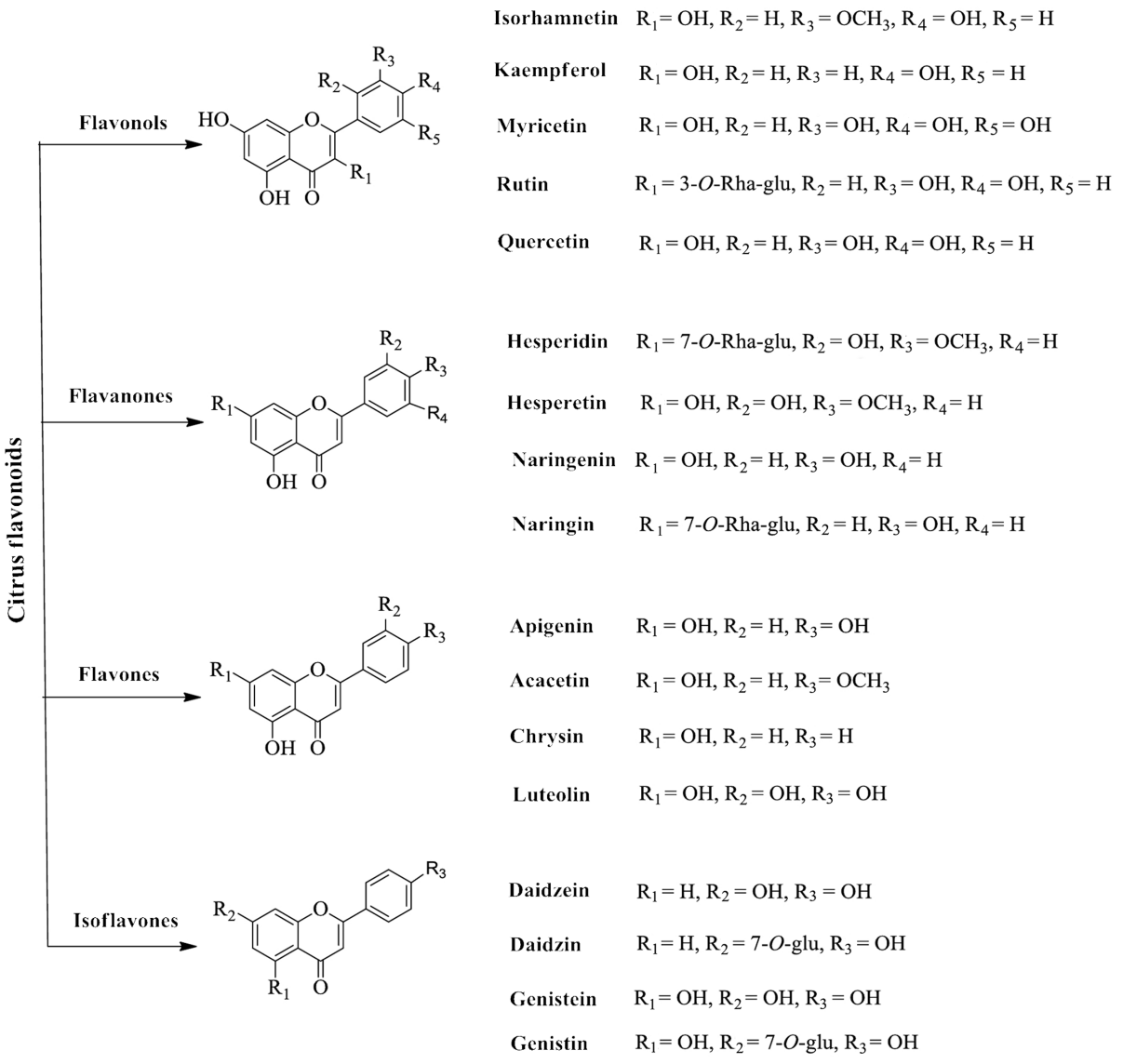



**Figure 4:** Classification of citrus flavonoids and chemical structures of the major flavonoids.

neohesperidose (rhamnosyl-α-1,2 glucose), and these are bitter in taste. On the other hand, rutinosides (flavanones, hesperidin, narirutin and didymin) consist a basic flavanone structure along with a disaccharide residue, e.g. rutinose (ramnosyl-α-1,6 glucose), and these are tasteless. The rutinosides are mainly found in oranges (*Citrus sinensis* L.), tangerines (*C. reticulate* L.) and lemons (*C. limon* L.) (Tripoli et al. 2007). Neohesperidosides are mainly found in the hybrids of grapefruit (*C. paradise* Macf.) and pomelo (*C. grandis* L.). Among the flavone aglycons, diosmetin is present in general citrus plants and luteolin is mainly found in lemons. Furthermore, lemons (*C. limon* L.) contain flavonols. The polymethoxyflavones, such as tangeretin, nobiletin, natsudaidain and heptamethoxyflavone are found abundantly in oranges, tangerines and lemons but are less abundant in some grapefruits (Manthey et al. 2001, González-Molina et al. 2010). Studies on the quantitative distribution of these flavonoids in citrus have shown that 7-*O*-glycosylflavanones are the most abundant flavonoids in all the citrus species (Benavente-Garcia et al. 1997).

Citrus flavonoids play important roles in physiological and ecological processes. These are also of commercial interest due to their wide applications in the food and pharmaceutical industries (Del Río et al. 2004). The flavanone glycosides are one of the foremost abundant flavonoids found in the edible part of the citrus fruits (Kawaii et al. 1999). The flavonoids greatly influence the quality of both the fresh fruit and the processed products.

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Hesperidin is a significant component contributing to cloudiness in lemon and orange juices (Nishimura et al. 1998), and naringin imparts a bitter taste to citrus fruits (Ranganna et al. 1983). On the other hand, polymethoxylated flavones are also important characteristic features of the citrus plants. These are minor components and mainly associated with the oil glands of the peel flavedo (Chen et al. 1997). Polymethoxylated flavones are reported to have many important bioactivities. In recent literatures on citrus flavonoids, elaborate studies on biological activities including anti-carcinogenic and anti-tumor activities have been reported (Benavente-García et al. 1997).

# 4 Sample preparation and characterization of citrus flavonoids

## 4.1 Extraction of flavonoids from citrus peels and pulps

Fresh, frozen or dried citrus samples can be used for the extraction of flavonoids. Usually, before extraction, the citrus samples are milled, ground and homogenized, which may be preceded by air-drying or freeze-drying. Generally, freeze-dried samples retain higher levels of flavonoids than the air-dried samples (Abascal et al. 2005). Solvent extraction is one of the most commonly used procedures to obtain extracts from plant materials due to their ease of use, efficiency and wide applicability. The yield depends on the nature of solvents used, e.g. solvent polarity, extraction time and temperature, sample-to-solvent ratio etc. The yield also depends on the chemical composition and physical characteristics of the samples. The main steps involved in the study of bioactive compounds present in citrus fruits are shown in Figure 5. The solubility of flavonoids is governed by the chemical nature of the sample, as well as the polarity of the solvents used. Methanol, ethanol, acetone, ethyl acetate and their combinations are commonly used for the extraction of flavonoids from citrus pulp and peel, often with different proportions of water. In particular, methanol has been found to be more efficient for the extraction of lower-molecular-weight flavonoids while the higher-molecular-weight flavonoids are better extracted with aqueous acetone (Labarbe et al. 1999).

In recent years, a number of efficient extraction methods have been developed, such as microwave extraction, ultrasound-assisted extractions, microwave-assisted extraction (Spigno and Faveri 2009), ultrasonic extraction (Londoño-Londoño et al. 2010) and techniques based on the use of compressed fluids, e.g. subcritical water extraction, accelerated solvent extraction (Kim et al. 2009, Khan et al. 2010). These techniques are also employed in the

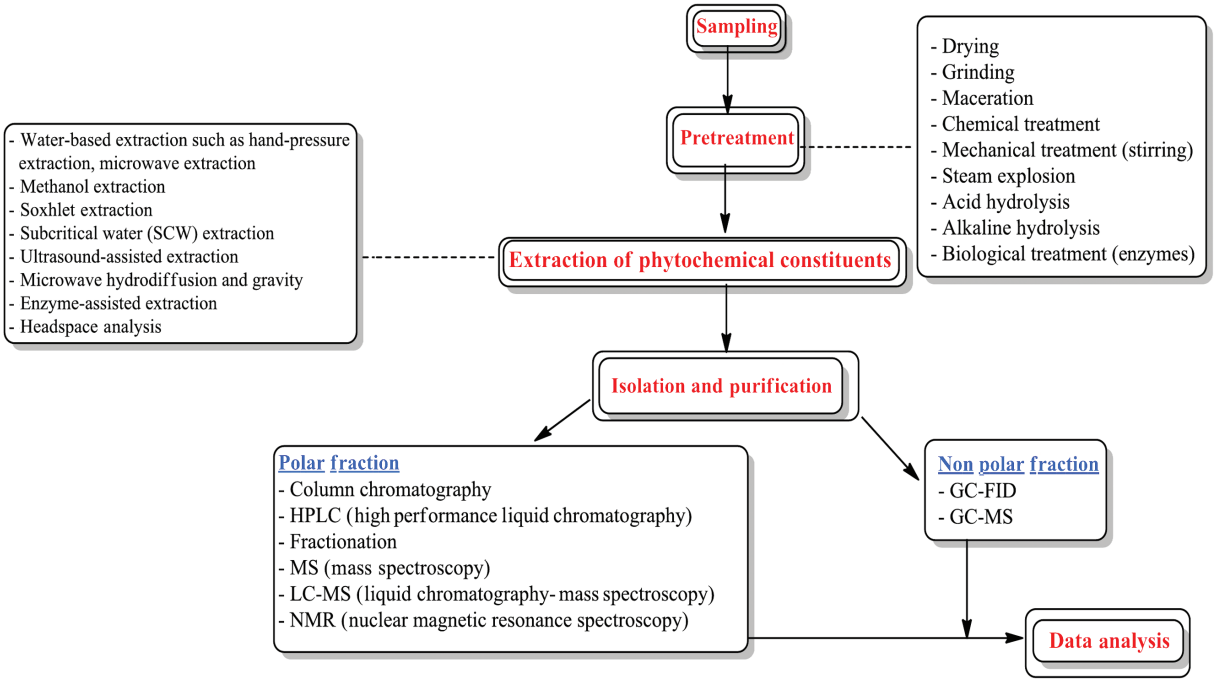



**Figure 5:** Main steps involved in the study of bioactive compounds present in citrus fruits.

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extraction of phenolic compounds from plant materials. Crude plant extracts usually contain large amounts of carbohydrates and/or lipoidal materials along with phenolics, rendering the concentration of the latter in the crude extract diluted or low. To concentrate and obtain Polyphenol-rich fractions before analysis, strategies including sequential extraction or liquid-liquid partitioning and/or solid-phase extraction (SPE) based on the polarity and acidity of solvents have been commonly used. In general, elimination of lipoidal materials can be achieved by washing the crude extract with non-polar solvents, such as hexane (Ramirez-Coronel et al. 2004), dichloromethane (Neergheen et al. 2006) or chloroform (Zhang et al. 2008). To remove polar non-phenolic compounds, such as sugars and organic acids, the SPE process is usually carried out. SPE is becoming popular since it is a rapid method of extraction and can be automated. Besides, it is a sensitive method, employs many varieties of sorbents with different cartridges and discs and is economical. C18 cartridges have been widely used for the separation of phenolic compounds. In this method, the aqueous sample is passed through preconditioned C18 cartridges, followed by washing with acidified water to remove sugar, organic acids and other water-soluble constituents. The polyphenols are then eluted with absolute methanol (Thimothe et al. 2007) or aqueous acetone (Ramirez-Coronel et al. 2004).

A typical method of extraction is as follows. A sample of 100 g of dried citrus, powdered from the peels and pulps, is defatted separately with n-hexane for 24 h followed by filtration and drying at room temperature. Soxhlet extractors are commonly used for the extraction process. Five hundred milliliters of 80% methanol is used as a solvent and placed in a 1-L round bottom flask fitted with a Soxhlet extractor. Extraction is continued for 12 h, and the resultant extracts are filtered and concentrated under reduced pressure to dryness using rotary evaporator with temperature not exceeding 40°C. The dry extracts are weighed and subjected to identification, isolation and purification procedures. Different analytical methods used for the identification of the isolated compounds are melting point measurement, thin layer chromatography (TLC), high-performance liquid chromatography (HPLC), Fourier transform infrared spectroscopy (FT-IR) and $^{1}$H nuclear magnetic resonance spectroscopy ($^{1}$H-NMR) analysis (Table 1).

**Table 1:** Analytical methods to determine different flavonoids from citrus fruits.

| Extraction and analysis methods | Type of compounds | Brief description |
|---|---|---|
| Heat treatment | Flavanone glycoside (FG) | Amounts of benzoic and cinnamic acids significantly increase; FG content and antioxidant capacity decreases after heat treatment (Xu and Chang 2007) |
| Vortexing with 1.2 M HCl in 80% methanol water (90°C, 3 h) | Flavonoids | Different edible tissues of fruits were frozen with liquid nitrogen and powdered in a mortar (Abeysinghe et al. 2007) |
| Soxhlet extraction (methanol, 80°C, 4 h); heat-reflux extraction (methanol, 80°C, 60 min) | Total and individual flavones | Identification by LC-DAD-ESI/MS (Li et al. 2002) |
| Methanol extraction and HPLC | Naringin, hesperidin, didymin, tangeretin and nobiletin | Analytical method for the simultaneous determination of five major flavonoids from citrus fruit by HPLC-PDA (Sun et al. 2010) |
| Subcritical water (SCW) extraction | Polymethoxylated flavones | Subcritical water extraction at 200°C for 60 min used for fast recovery of phenolic compounds (Kim et al. 2009) |
| Stirring with methanol (25°C, 12 h) hydrodistillation | Total phenols and flavonoids content | Polar compounds will be isolated with methanol and hydrodistillation for volatile compounds (Guimaraes et al. 2010) |
| Ultrasound-assisted extraction | Flavanone glycosides | Optimum particle size of 2 $cm^2$ is considered best for ultrasonic waves and appeared very effective in comparison to conventional procedure (Khan et al. 2010) |
| Ultrasound-assisted extraction method with HPLC/DAD-MS analysis | Hesperidin and neohesperidin, diosmin, nobiletin, tangeritin | Ultrasound-assisted extraction, frequency of 60 kHz, extraction time of 30 min, temperature of 40°C, citrus peel/water ratio (g/ml) 1/10, $Ca(OH)_2$ as basifying agent and water as solvent (Londoño-Londoño et al. 2010) |
| Optimize HPLC method | Diosmin, hesperidin and naringin | A reversed-phase HPLC method for the simultaneous determination was simple, specific, precise and accurate and is successfully used for the quantitative analysis (Ferhat et al. 2007) |
| LC-DAD and LC-MS | Flavonoids (isoquercitrin) | Rapid sample preparation and flavonoid determination can be used in quality assurance routine analysis for adulteration detection in citrus concentrates and juices (Bilbao et al. 2007) |

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## 4.2 Identification of citrus flavonoids by TLC

For the identification of flavonoids on TLC, a small amount of solvent extract (1 mg dried powder dissolved in 1 ml solvent) of fruit peels and pulps are manually applied (1 mg/ml) on TLC plates along with standard samples using capillary tubes. Different solvent combination systems, such as hexane:*n*-butanol, ethyl acetate:hexane, butanol:water, chloroform:methanol etc., have been used for this purpose. Water is used for the detection of flavonoids in different fruit parts. Detection is done by using UV light at $\lambda = 254$ nm and 366 nm. The flavonoids standards used for comparison are neohesperidin, naringin, hesperidin, narirutin, tangeretin and nobiletin (Garcia et al. 1993).

## 4.3 UV spectra of citrus flavonoids

The UV spectra of flavones and related glycosides generally show two strong absorption peaks commonly referred to as band I ($\lambda = 300–380$ nm) and band II ($\lambda = 240–280$ nm). Band I is associated with the presence of a B-ring cinnamoyl system. Band II absorption is due to an A-ring benzoyl system. Substitutions on the A or B ring may produce hypsochromic or bathocromic shifts in the absorption spectrum, which are useful for identifying the corresponding flavonoids structures (Mabry et al. 1970). The photodiode array (PDA)/UV spectra of naringin, hesperidin, didymin, tangeretin and nobiletin display characteristic peaks in the range $\lambda = 210–400$ nm. Naringin, hesperidin and didymin have similar spectra with high absorptions at about $\lambda = 210–227$ nm and $\lambda = 283–285$ nm. However, tangeretin shows maximum responses at wavelengths $\lambda = 210$, 250, 270 and 334 nm. Nobiletin shows maximum responses at the wavelengths of $\lambda = 210$, 271 and 324 nm (Sun et al. 2010).

## 4.4 HPLC spectra of citrus flavonoids

HPLC is one of the most widely used characterization techniques for the analysis of flavonoids (Hvattum 2002). HPLC generally does not require preliminary derivatization in the case of flavonoids. The C-8 or C-18 columns are frequently used in reversed-phase chromatography for the separation of citrus flavonoids (Penazzi et al. 1995) with polar mobile phases, such as methanol, acetonitrile, tetrahydrofuran or acid solutions (Merken and Beecher 2000). Gradient elution is the best choice for the separation of different flavonoids present in the citrus fruit extract (Khokhar and Magnusdottir 2002). Under normal reversed-phase condition, generally, the more polar compounds are eluted first. Thus, diglycosides are eluted in the beginning followed by monoglycosides and aglycones in the end. The classes of flavonoids characteristic to particular citrus species exhibit their maximum absorption in specific wavelength ranges, *viz*. flavanones ($\lambda = 280–290$ nm), flavones ($\lambda = 304–350$ nm) and flavonols ($\lambda = 352–385$ nm). A DAD chromatogram recorded for orange extract at $\lambda = 330$ nm displaying peak positions of different flavonoids is presented in Figure 6.

## 4.5 Liquid chromatographic/mass spectrometric analysis

Mass spectrometry is an analytical technique capable of identifying mass of a molecule along with its fragments. In

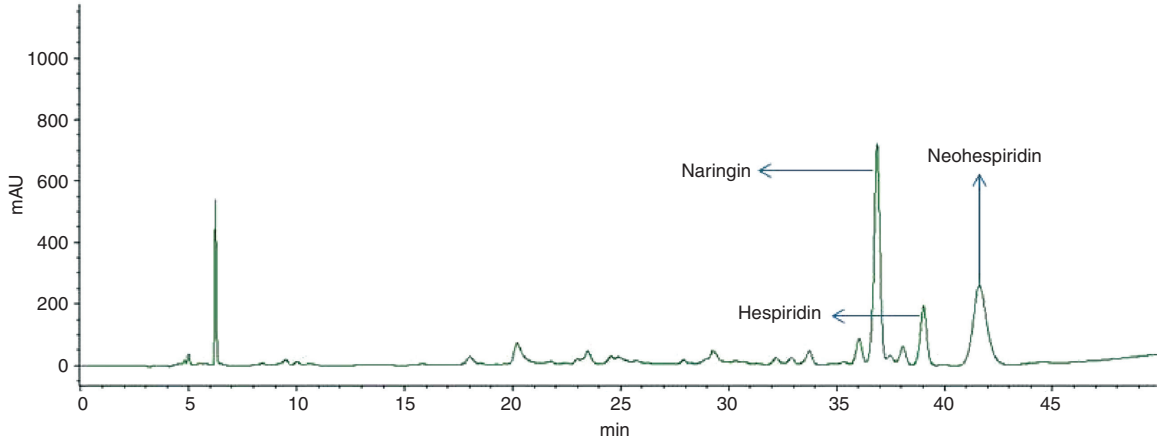



**Figure 6:** HPLC chromatograms of orange extract at 330 nm.

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a typical mass spectrometer, electric and magnetic fields are generated inside the instrument which fragments a big molecule into smaller ones and ions are separated according to their mass-to charge ratio ($m/z$). A typical mass spectrum of an organic compound consists of a plot of ion abundance versus its $m/z$ ratio. A molecular ion is usually $[M+H]^+$ or $[M-H]^-$; thus, by knowing its $m/z$, the molecular mass of the compound can be deduced. There are various types of ionization sources that can be used as the interface between liquid chromatography (LC) and mass spectrometer (MS), such as electrospray ionization (ESI), atmospheric pressure chemical ionization (APCI), matrix-assisted laser desorption ionization (MALDI) etc. The following ionizing sources improve MS sensitivity and performance for high throughput analysis of biological samples. For the analysis of small metabolites, such as bioflavonoids, the ESI/MS and APCI/MS offer excellent mass range and sensitivity (Prasain and Barnes 2007). Recently, a new ionizing method for LC/MS, atmospheric pressure photoionization, has been introduced, which is based on charge transfer from dopant molecules (e.g. toluene) to the analytes. The dopant molecules are ionized using 10-eV photons produced by a vacuum UV lamp. Karas et al. (1987) introduced another ionization source, i.e. MALDI, for the analysis of nonvolatile compounds. Although MALDI/TOF/MS is a powerful and well-known tool for the analysis of a wide range of biomolecules, such as bioflavonoids, peptides and proteins, its potential in food analysis is explored only recently. MS techniques are useful in determining the molecules possessing health benefits, such as flavonoids and other polyphenols. HPLC paired with UV photodiode array and electrospray ionization tandem mass spectrometry detectors (HPLC-PDA-ESI/MS) are used for the determination of phenolic compounds and their proposed structures are further confirmed by NMR spectroscopy (Piccinelli et al. 2008).

Purification step is crucial for the analysis of flavonoids using LC/MS, as methanolic extracts are generally complex mixtures. Gas chromatography (GC)/MS is not widely used in the flavonoid analysis owing to the limited volatility of flavonoid glycosides. In addition, it requires derivatization, which makes the analysis process more time-consuming, and the fragmentation patterns of the derivatives are often difficult to interpret. Nowadays, the coupling of atmospheric pressure ionization (API) sources with LC/MS is emerging as one of the highly efficient as well as popular techniques for online analysis of flavonoids. LC/MS provides the molecular mass of the different constituents in flavonoids, but it is rarely used for full-structure characterization. Knowledge of LC/MS/MS and MS/MS analysis can be pursued for further structure characterization. C8-and C18-RP columns are generally used in LC/MS, as these offer better retention and thereby facilitating efficient separation of different flavonoids present in the citrus extract (Constant and Beecher 1995). Buffers, such as formic, acetic and trifluoroacetic acid and ammonium acetate and formate, are volatile and thus compatible with LC/MS systems. In the LC/MS spectrum, the peak with highest $m/z$ ratio does not always correspond to the molecular ion species or base peak (e.g. $[M+H]^+$ and $[M-H]^-$) in the positive and negative mode, respectively. In some cases, adduct species are formed with solvent and/or acid molecules, e.g. $[M+Na]^+$ and appear as the base peak in the spectrum (Barnes et al. 1994). The adduct species result in the formation of very few fragment ions with a low relative abundance and are very difficult to detect. When the peak corresponding to sodium adduct ion or $[M+Na]^+$ dominates in the spectrum, ammonium acetate is added at a certain concentration (~0.1%) to suppress the signal. Addition of ammonium acetate has an advantage – the ammonia formed during the process is easily evaporated in the ESI source, leading to an increase in the $[M+H]^+$ signal. In addition, molecular complexes, e.g. $[2M+H]^+$ or $[2M-H]^-$ can also be generated (Sägesser and Deinzer 1996). An increase in voltage reduces the probability as well as incidences of both adduct and complex formation (Barnes et al. 1994). The first-order mass spectrum obtained from positive ion mode contains more structural information than that from negative ion mode, which is useful to identify known compounds. The combined use of both ionization modes helps in extracting more information for the molecular mass determination. This is useful especially in the case of minor compounds where the noise level is much higher. Similarly to HPLC, in LC/MS, the more polar compounds are eluted first. Thus, retention times are inversely correlated with increasing glycosylation, whereas acylation, methylation or prenylation have opposite effects. The position of glycosylation (Harborne and Boardley 1984) or methylation (Greenham et al. 1995) significantly influences the retention time in LC/MS. Flavanones are eluted first, followed by flavonols and flavones for the flavonoids compounds with equivalent substitution patterns. For isomeric compounds that differ in the structure of the saccharide residues, rutinosides are eluted prior to neohesperidosides; galactosides prior to glucosides (Robards et al. 1997); glucosides prior to arabinosides and arabinosides prior to rhamnosides (Schieber et al. 2002). It is therefore important to consider the linkage positions that have significant effects on the retention. The MS and MS/MS spectra of flavonoid glycosides have typical patterns, which depend mainly on the number or nature of the bound saccharides and their *C*- or

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*O*-glycosidic linkages (Ferreres et al. 2004). In contrast to *O*-glycosides, for which fragmentation can easily occur at the loss of sugar moiety, the fragmentations in *C*-glycosides occur preferably at the glycidic moiety (Stobiecki 2000). The type and number of fragments formed during fragmentation indicate the linkage positions between the sugar and the aglycone. At low collision energies, the main fragments formed from flavone-*C*-glycosides are observed due to loss of water moiety ($[M+H\text{-}nH_2O]^+$) and loss of glucosidic methylol group as formaldehyde ($[M+H\text{-}CH_2O\text{-}2H_2O]^+$) (Abad-García et al. 2009). Apigenin-6,8-di-*C*-glucoside is represented by a positively charged molecular ion ($[M+H]^+$) at *m*/*z* 595, which yields secondary fragments at *m*/*z* 577 ($[M+H\text{-}H_2O]^+$), 559 ($[M+H\text{-}2H_2O]^+$), 529 ($[M+H\text{-}CH_2O\text{-}2H_2O]^+$) and 523 ($[M+H\text{-}4H_2O]^+$), respectively (Roowi and Crozier 2011). Diosmetin-6,8-di-*C*-glucoside and chysoeriol-6,8-di-*C*-glucoside show the following fragmentation patterns: $[M+H]^+$ ion at *m*/*z* 625, with the presence of fragments assigned to *m*/*z* 607 ($[M+H\text{-}H_2O]^+$), 589 ($[M+H\text{-}2H_2O]^+$), 571 ($[M+H\text{-}3H_2O]^+$) and 541 ($[M+H\text{-}CH_2O\text{-}3H_2O]^+$) ions (Zheng et al. 2009).

The ESI/MS spectrum of an *O*-disaccharide-substituted flavanone, i.e. hesperidin (hesperetin 7-*O*-rutinoside), recorded in negative mode (Gattuso et al. 2007), displays a prominent fragment peak at *m*/*z* 463, generated by the loss of one sugar unit (rhamnose) from the pseudo-molecular ion $[M-H]^-$ (*m*/*z* 609). The subsequent loss of the second sugar unit (glucose) generates the ion *m*/*z* 301, which is easily assignable to aglycone. The MS spectrum of hesperidin recorded in the positive mode shows a different pattern of fragmentation. The pseudo-molecular ion $[M+H]^+$ undergoes a partial rearrangement along the fragmentation pathway, which is peculiar for such types of compounds. This leads to the loss of the "internal" sugar moieties of the disaccharide (glucose, in the example described) demonstrated by the presence of $[M+H\text{-}162]^+$ ions (Ma et al. 2000). The fragmentation, however, is concomitant with the expected pathway, i.e. the loss of rhamnose moiety. MS/MS experiments are very useful for identifying aglycones present in a given flavonoid molecule (Gattuso et al. 2007). Fragmentation pattern analysis is thus a highly diagnostic approach, and it is possible to obtain a precise structure elucidation of the aglycones by comparison with the literature data (Fabre et al. 2001).

The composition of polymethoxylated flavones can significantly differ in different citrus species and varieties (Bonaccorsi et al. 2005). Therefore, a rapid and unambiguous assignment of the constituents in their extracts is necessary. Reports on LC/MS studies performed on the flavonoids from citrus (Aturki et al. 2004) indicate that the polymethoxylated flavones can be normally identified by comparing the MS and UV spectra generated by the respective instruments with commercially available standard materials or previously isolated reference materials. The spectral analysis and identification of the flavonoids present in the citrus fruits by LC/UV, LC/MS and MS/MS data have been recorded in Table 2. The identification of citrus flavonoids is performed with UV/Vis spectra, and the corresponding retention times are compared with chemical standards (Chinapongtitiwat et al. 2013). Because of the limitation of available commercial standards, only a few components, such as hesperidin and naringin, found in the citrus fruits can be analyzed and determined quantitatively. UHPLC coupled with high-resolution mass spectrometry (UHPLC/HRMS) is one of the best analytical techniques, which combines the high separation efficiency of UHPLC and the excellent structure identification capability of MS. UHPLC/HRMS is used for the separation and identification of flavonoids found in different parts of citrus fruits, including juices, peels and pulp without prior purification (Barreca et al. 2013). Using high-resolution mass spectrometers, the accurate mass of parent and fragment ions can be measured. Moreover, tandem mass spectrometry is able to provide valuable fragment information required to deduce the chemical structures of unknown compounds and does not require standards (Dunn et al. 2013). Based on HPLC/DAD/ESI/MS/MS, Abad-Garcia et al. (2009) developed a general strategy for the characterization of phenolic compounds in citrus juices by investigating the fragmentation pattern of 72 standards.

The LC/NMR technique is employed as a complementary tool which speeds up the evaluation process of the compounds of interest by providing structural information in the form of NMR spectra. A rising number of scientific reports and publications on LC/NMR studies on natural extracts and ongoing technical developments indicate the increasing importance of this analytical tool (Exarchou et al. 2005). The NMR technique is extensively employed to identify the chemical structures of isolated citrus flavonoids. The chemical structures of purified compounds are always identified by using NMR technique combined with mass spectrometry. For example, 10 PMFs isolated from the peel of mandarin were identified by spectral analysis using MS and NMR (Manach et al. 2003). Using HPLC/MS and NMR technique, two di-*C*-glycosyl flavones, a series of flavones, flavanone 7-*O*-neohesperidosides and two methoxyflavones (nobiletin and tangeretin), commonly present in citrus, were identified in *Citrus aurantium* var. *amara* L. peel (Mencherini et al. 2013). Also, derivatives of coumarins and furocoumarins in distilled lemon peel oils are analyzed using LC/MS and LC/NMR (Sommer et al. 2003). HPLC

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**Table 2:** Spectral analysis and identification of flavonoids present in citrus fruits by LC/UV, LC/MS and MS/MS data (He et al. 1997, Brito et al. 2014, Salerno et al. 2016).

| Compounds | HPLC DAD λ Max (nm) | $[M-H]^-$/ $[M+H]^+$ (*m*/*z*) | MS2 ions (*m*/*z*) | MSn ions (*m*/*z*) |
|---|---|---|---|---|
| Flavones | | | | |
| Apigenin 7-*O*-neohesperidoside-6-*C*-glucoside | 268, 334 | 739 | 431 (M-neohesperidose) | 311 (431–120) |
| Diosmetin 6,8-di-*C*-glucoside | 250, 268, 342 | 623/625 | 503 (M-120) | 383, 312 |
| Diosmetin 8-*C*-glucoside | 250, 268, 342 | 461/463 | 341/343 (M-120) | 298 |
| Luteolin 7-*O*-rutinoside | 254, 267 | 593/595 | 285/287 (luteolin) | 241, 175 |
| Luteolin | 254, 267 | 285/287 | 269 (M-16) | 243, 241, 217 |
| Chrysoeriol 6,8-di-*C*-glucoside | | 623/625 | 503 (M-120) | 383, 312 |
| Diosmetin-7-*O*-rutinoside | 250, 268, 342 | 607/609 | 563 299 (diosmetin) | 284 |
| Apigenin 6,8-di-*C*-glucoside | 268, 334 | 593/595 | 512 | 473 (M-120), 353 (M-240), 297 |
| Chrysoeriol 6,8-di-*C*-glucoside | | 623/625 | 503 (M-120) | 383, 312 |
| Luteolin 6,8-di-*C*-glucoside | 254, 267 | 609/611 | 489 (M-120) | 369 (M-240) |
| Flavanones | | | | |
| Hesperidin | 285, 330 | 611 | 371 | |
| Neohesperidin | 285, 330 | 611 | 371 | |
| Naringin | 285, 330 | 581 | 356 | |
| Neoeriocitrin | | 595 | 287.05, 459.11 | |
| Flavonols | | | | |
| Rutin | 254, 354 | 609/611 | 301 (M-rutinose) | 179, 151 |
| Quercetin | 254, 354 | 301 | 179, 151 | 179, 151 |

separation has significantly assisted the subsequent detection and characterization of the polymethoxylated flavones by MS and NMR.

## 4.6 NMR spectra of citrus extracts

Metabolites differ in their properties regarding their polarity, chemical behavior, stability and concentration, which make the analysis in one single experiment extremely difficult. Under certain conditions, NMR can be employed to analyze all the metabolites present in the citrus extract instead of conducting quantitative and qualitative analysis of the same. NMR is a very suitable and fast method for simultaneous detection of diverse groups of secondary metabolites (flavonoids, alkaloids, terpenoids and so on). However, it is not possible to analyze the primary metabolites, *viz*. sugars, organic acids, amino acids etc., quantitatively. NMR spectrum is not only used for the structure elucidation, but also for the quantitative estimation in terms of concentration. In NMR spectrum, the signal intensity is proportional to the molar concentration of the compound. This enables direct comparison of concentrations of different compounds possible, and the prerequisite of plotting calibration curves of each individual compound is not mandatory. Furthermore, NMR is a very useful technique for metabolomics analysis, but it has several limitations, *viz*. low sensitivity, requirement of large amounts of samples compared to other analytical methods, such as HPLC and LC/MS, and signals of different metabolites with same nature overlap and make the spectrum more complicated. Typically, $^1$H-NMR spectra contain hundreds of overlapping signals that may complicate signal identification and accurate peak integration. This overlapping can be resolved to a great extent by using two-dimensional (2D) NMR. The latter has a much better resolution than the former. In most cases, $^1$H-NMR is sufficient to generate metabolomic data of the sample in relatively shorter time (~5–10 min for 64–128 scans). A typical $^1$H-NMR spectrum of citrus flavonoids contains several signals, and it reflects the amount of all flavonoids in the extract. NMR signals are directly proportional to the molar concentration of the particular compound independent of its characteristic. The absolute concentration of the flavonoids can be calculated by comparing the peak intensities with the internal standard. The internal standard, 1,3-(tri-methylsilyl) propionic-2,2,3,3-d4 acid is a widely employed reference for the calibration of NMR shifts and can also serve as a reference (internal standard) in the quantification of metabolites. Depending on the solvent, different standards can be used as reviewed by (Pauli et al. 2005). Spectra of authentic substances from the standard data include common primary metabolites and secondary metabolites, such as phenolic compounds (benzoic acid, salicylic acid, gallic acid, cinnamic acid derivatives and so on) measured under standard conditions as described

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by a set protocol. A library with 1-D and 2-DNMR spectra of more than 300 compounds is available in the database for identification and comparison. Currently, more than 20,000 spectra of plant extracts are available, allowing direct comparison of the data from new experiments with those of previous experiments. Additionally, direct coupling of the NMR instrument to a LC/UV/MS system results in a unique and a very powerful technique. However, the resultant technique becomes an expensive means for the identification of the unknown and complicated flavonoid compounds of plant origin (Lommen et al. 2000).

## 5 Biological activities of citrus flavonoids and application

Citrus flavonoids exhibit a range of biological activities, e.g. anticarcinogenic, lypolytic, analgesic, antitumor (Benavente-García et al. 1997, Chen et al. 1997). Flavonoids are one of the most prominent cancer-preventing agents (Stavric 1994). Diosmin and hesperidin have been observed to influence the multiplicity of neoplasm in the large intestine of male F344 rats (Tanaka et al. 1997). Several polymethoxylated flavones show activity toward differentiation in human acute promyelocytic leukemia cells (HL-60) (Kawaii et al. 1999). Luteolin and natsudaidain demonstrate strong activity toward non-proliferation of several cancer cell lines, whereas they showed weak activity against the normal human cell lines (Kawaii et al. 1999). Naringin has been found to lower the total cholesterol and low-density lipoprotein cholesterol levels in blood plasma (Jung et al. 2003). Hesperetin and its metabolites have demonstrated significant reduction in the total cholesterol and triglyceride concentrations in the blood plasma (Kim et al. 2003). Hesperidin and diosmin, both alone and in combination, act as a chemopreventive agent against colon carcinogenesis (Tanaka et al. 1997). The polymethoxylated flavone, nobiletin, has been found effective in the down-regulation of the production of promatrix metalloproteinase and interfering with the proliferation of synovial fibroblasts (Ishiwa et al. 2000). Tangeretin has been reported to suppress malignant tumor invasion and metastasis (Bracke et al. 1994).

Flavonoids are powerful antioxidants against free radicals and act as "radical-scavengers". The activity is attributed to their ability of releasing protons. The phenolic groups of flavonoids serve as a source of H-atoms which are readily available. Subsequent formation of radicals can be delocalized over the flavonoid structure (Di Majo et al. 2005). The chemical nature of the flavonoids depends on structural class, degree of hydroxylation, other substitutions and conjugations and degree of polymerisation (Calabro et al. 2004). Polyphenolic compounds present in *Citrus limetta* peels have demonstrated effects on the carbohydrate metabolism by inhibiting the activity of enzymes, namely α-glucosidase and α-amylase, which are responsible for carbohydrate digestion. It also helps in preventing chronic hyperglycemia or type 2 diabetes mellitus (Johnston et al. 2005). Flavanones and polymethoxyflavone have been reported for estrogen effects. They are believed to be useful in hormone replacement therapy in women, especially in the post-menopausal stage. The untreated cases are vulnerable toward further complications of osteoporosis, coronary diseases and deficiency of calcium, resulting in the reduction of bone density, abnormal blood cholesterol profile, anxiety, thyroid functioning, hot flashes and insomnia. Conventional pharmacological treatments for hormone replacement in women have shown a number of side effects, e.g. increased risks of cardiovascular malfunctioning, stroke, symptoms of breast cancer, thyroid malfunctioning, clotting of blood etc. (Mahato et al. 2017). Macromolecules present in our body fluid play an important role in the aggregation of calcium oxalate crystals leading to stone formation or urolithiasis. Calcium oxalate crystals are sparingly soluble or insoluble in water and gradually accumulate in the form of stones leading to gall bladder malfunctioning and kidney failure. In many cases, the stone formation reoccurs even after treatment or removal and remains a big challenge for the medical experts. Citric acid derived from citrus resources helps in converting these oxalates into citrates, which are relatively more soluble in water and can be excreted through urine. In recent years, incorporation of phytotherapy based on citrus-derived bioactive compounds, e.g. naringenine, hesperidin, rutin, into modern medicine has started gaining interest. Some of the recent scientific reports on the effects and health benefits of citrus-derived phytochemicals are described in Table 3 (Mahato et al. 2017).

## 6 Future perspectives

The demand for the extraction of bioactive compounds from fruit resources encourages continuous search for a more convenient, non-destructive, safe and economical methods. The sustainable techniques for the extraction of bioactive compounds depend on its nature, class and source. The most challenging aspect in the field of extraction of bioactive compounds is improvisation and implementation of laboratory techniques at industrial scale for commercial benefits.

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**Table 3:** Citrus waste-derived substances for health benefits and pharma-/nutraceutical applications.

| Active compound(s) | Effect on health | Animal model | References |
|---|---|---|---|
| Naringenin 7-*O*-β-D-glucoside | Decreases blood glucose and improves lipid oxidation | Streptozotocin-induced diabetic rats | Choi et al. 1991 |
| Hesperidin, narirutin | Anti-aging effect | *In vivo* (mice) | Tokunaga et al. 2016 |
| Flavonoids, polymethoxyflavonoids | Hepatoprotective and immunosuppressive effects | *In vivo* (mice) | Pantsulaia et al. 2014 |
| Naringenin and naringenin-7-*O*-cetyl ether | Lowers plasma cholesterol concentration and hepatic cholesterol content by inhibiting the activities of 3-hydroxy-3-methylglutaryl coenzyme A (HMG-CoA) reductase and acyl coenzyme A cholesterol *O*-acyltransferase (ACAT); decreases plasma TG, TC; increases HDL-C; decreases hepatic TG and TC | Male Sprague-Dawley rats fed on high-cholesterol diets | Lee et al. 1999 |
| Phenolic compounds and flavonoids | Antimicrobial effect against dental caries bacteria *Streptococcus mutans* and *Lactobaccillus acidophilus* | *In vitro* (agar well diffusion method) | Shetty et al. 2016 |
| Naringenin | Increases activity and expression of genes involved in hepatic fatty acid oxidation by increasing the activity of hepatic cyanide-insensitive palmitoyl-CoA; increase in the oxidation rate by upregulation of gene expression of enzymes involved in peroxisomal fatty acid oxidation; upregulation of gene expression of enzymes, such as carnitine octanoyltransferase, acyl-CoA oxidase, bifunctional enzyme and 3-ketoacyl-CoA thiolase etc | Male ICR mice | Huong et al. 2006 |
| 8-Geranyloxypsolaren; phloroglucinol 1-β-D-glucopyranoside (phlorin) | Antimicrobial effect against dental caries bacteria, viz. *Streptococcus mutans* ATCC7270, *Prevotella intermedia*, *Porphyromonasgingivalis* 381 | *In vitro* | Miyake and Hiramitsu 2011 |
| Flavonoids-naringin, narirutin, neohesperidin | Cardioprotective activity against cyclophosphamide- and doxorubicin (DOX)-induced cardiac toxicity in rats | *In vivo* (adult male Wistar rats) | Baniya et al. 2015 |
| Flavonoids (naringin) | Ameliorating effect on DOX-induced cardiomyopathy | *In vivo* (adult male Sprague-Dawley rats) | Saalu et al. 2009 |
| Red grapefruit | Improves plasma lipids and blood pressure by significantly decreasing total cholesterol and low-density lipoprotein, but does not significantly decrease body weight, lipids, or blood pressure as compared with the control condition | Overweight and obese men and premenopausal women | Dow et al. 2012 |
| Naringenin | (2*S*)-Naringenin suppresses the DNA synthesis and proliferation *via* a $G_0/G_1$ cell cycle arrest; may be useful for individuals at high risk of thrombotic or cardiovascular disease | VSMCs isolated from Sprague-Dawley rats | Lee et al. 2012 |
| Flavonoids (naringin, hesperidin) | Hepatoprotective effects; improves lipid and bone metabolism | *In vivo* (ovariectomized [OVX] rats – animal model of post-menopausal osteoporosis) | Lim et al. 2014 |
| Naringenin | Reduces plaque progression; improves dyslipidemia and biomarkers of endothelial dysfunction; changes the gene expression that may lead to preservation of the vascular wall (studied by aorta transcriptomic analysis) | *C57BL/6* mice fed with high-fat and high-cholesterol diet | Chanet et al. 2012 |
| Flavanone and polymethoxyflavone | Modulation of blood cholesterol and bone density | *In vivo* (OVX rats) | Adelina et al. 2015 |
| Orange juice or hesperidin | Improves endothelium-dependent vasodilation diastolic blood pressure | Healthy men | Morand et al. 2011 |

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**Table 3** (continued)

| Active compound(s) | Effect on health | Animal model | References |
|---|---|---|---|
| Flavonoids and phenolic acids | Anti-ulcerogenic and antioxidant activity | *In vivo* (Wistar rats; water immersion and hypothermic restraint stress models) | Sood et al. 2010 |
| Sinetrol (citrus extract) | Sinetrol is a potent inhibitor of cAMP-phosphodiesterase compared to other purified compounds (narangin, caffeine); it also suggests that sinetrol has lipolytic effect and results in an increased fatty acid release; prevents obesity by decreasing BMI | Human adipocytes | Dallas et al. 2008 |
| Flavonoids and phenolic acids | Anti-ulcer activity | *In vivo* (mice) | Nagaraju et al. 2012 |
| Polyphenols, flavonoids | Anti-obesity effect | *In vivo* (mice) | Kim et al. 2016 |
| Flavonoids | Antibacterial effect | *In vitro* (antimicrobial assay) | Kumar et al. 2011 |
| Flavonoids, polymethoxyflavones (PMFs) | Anticancer activity (skin, colon, prostate) | *In vivo* (female ICR mice) | Suzawa et al. 2014 |
| flavanone rutinoside and flavanone glycoside | Antioxidant and anti-inflammatory activity | *In vitro* (RAW 264.7 cells) | Chang et al. 2016 |
| Hesperidin and narirutin | Proangiogenic effects; improves blood circulation | *In vitro* (human umbilical vein endothelial cells) | Lee et al. 2016 |
| Bioflavonoid naringenin | Cholesterol regulation | *In vivo* (rats) | Shin et al. 1999 |
| Bioflavonoid naringenin | Hepatic cholesterol regulation | *In vivo* (Sprague-Dawley rats) | Kim et al. 2006 |
| Polyphenol extract of citrus fruits | Circumference oxidative stress is lowered by the reduction of malondialdehyde and the increase in superoxide dismutase and glutathione results in the reduction of abdominal fat, waist and hip fats; no effect on kidney, liver and lipid panels | Overweight humans | Dallas et al. 2014 |
| Phenolics and flavonoids | Anti-aging effect | *In vitro* (antioxidant and anti-enzyme assay) | Vinita et al. 2016 |
| Bioflavonoids | Inhibition of stone-forming proteins | *In vivo* (rats) | Sridharan et al. 2016 |
| Auraptene | Anti-inflammatory effect | *In vivo* (mice) | Okuyama et al. 2014 |
| Nobiletin | Enhances learning and memory | *In vivo* (mice) | Kawahata et al. 2013 |
| Nobiletin | Increases hepatic and peripheral insulin sensitivity and glucose tolerance; dramatically attenuates atherosclerosis in the aortic sinus; reduces glucose and insulin; improves glucose tolerance | *Ldlr* – / – mice fed with high-fat diet | Mulvihill et al. 2011 |



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The combination of different technologies to enhance the extraction of bioactive compounds has not been completely exploited, for example microwave with pressurized fluid extraction. Most of the extraction techniques are already installed in the industries, but the use of these techniques for the extraction of bioactive compounds from food wastes, such as citrus waste, has not yet been established. The extraction yield can vary depending upon different factors, such as solubility, selectivity, kinetic parameters and mass transfer. To optimize these parameters for maximum yield, theoretical simulations and mathematical modeling are also required to be incorporated. The other challenges include extraction of bioactive compounds present in minor quantities in the byproduct wastes and their probable utilization in commercial applications.

Summing up all the above considerations, the future investigations should also incorporate integral studies on the development of recovery protocols, findings on specific applications to secure possible industrial exploitation and sustainability of the final product. Finally, it is very important and necessary to raise awareness among the consumers regarding advantages of these technologies in order to create a new class of functional foods based on natural nutraceuticals as alternative to the commonly used synthetic pharmaceutical compounds. In this review, we enlisted the important research reports on the biological activities of citrus flavonoids. The most challenging issue in the research on the bioavailability of crude extract as well as purified compounds is the lack of consistency in the studies *in vivo*, including both animals and humans. One of the most probable reasons might be the differences in the enteric microflora population present in the test models, giving rise to different interindividual variability in absorption and pharmacokinetic parameters. The primary mechanism(s) of action is not well understood and the differences observed in the bioavailability results have made the selection of optimal doses difficult. Moreover, in depth studies are needed to be carried out to further explain and validate the molecular and cellular mechanisms of the biological activities of citrus flavonoids as well as its metabolites in the animal body.

An interesting area worth of further exploration is the biological activities of citrus flavonoids mixture and synergic effects of different bioactive compounds present in citrus. As the diet we consume in our daily life is a complex food matrix containing several bioactive components, understanding this aspect will impart significant role in programming healthy food habits. Simultaneously, further clinical studies in different populations using purified compounds should be performed to clarify the beneficial effects of citrus flavonoids in humans. Also, employing genetic models with a combination of both gene expression and protein profiling technologies, combined with rigorous biochemical analyses, would be a promising approach for deciphering both classic and novel mechanisms of action of citrus flavonoids in animals and humans.

## 7 Summary

Citrus flavonoids are powerful bioactive phytochemicals and possess diversified physiological effects. The physiologically relevant effects of citrus foods are most likely due to the flavonoid molecules. The present article includes the description of native citrus species and the main commercial varieties cultivated in tropical and subtropical countries and the increasing concern on environmental pollution because of citrus wastes. Important bioactive compounds, especially flavonoids, are extracted by a number of modern extraction methods. The article summarizes different scientific reports on the identification and characterization techniques employed in developing flavonoids profile and their relative merits. Three structural groups are important for the evaluation of their antioxidant capacity: the *ortho*-dihydroxy (catechol) structure in the B-ring, the 2,3-double bond in conjugation with a 4-oxo function and the presence of both 3-(*a*)-and 5-(*b*)-hydroxyl groups. Flavonoids are also known as ‘‘nutraceutical substances”. These possess anticancer, antitumor, analgesic, lypolytic, anti-atherogenic, antimicrobial and anti-inflammatory and therapeutic properties. Therefore, increasing awareness toward a healthy lifestyle has incorporated citrus as an essential part of diet and cuisine.

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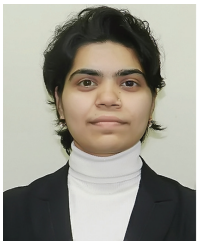
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